

Chapter 01: Real Numbers

① Rational and Irrational Numbers:

Definition:

$$\mathbb{Q} = \left\{ x \in \mathbb{R} / x = \frac{p}{q}, p \in \mathbb{Z} \text{ and } q \in \mathbb{Z}^+ \right. \\ \left. \text{such that } \text{PGCD}(p, q) = 1 \right\}$$

Propositions:

P₁: The sum of two Rationals is a Rational

Proof: $x_1 = \frac{p_1}{q_1} / p_1 \in \mathbb{Z}, q_1 \in \mathbb{Z}^+$

$$x_2 = \frac{p_2}{q_2} / p_2 \in \mathbb{Z}, q_2 \in \mathbb{Z}^+$$

$$x_1 + x_2 = \frac{p'}{q'} \quad \left\{ \begin{array}{l} p' = p_1 q_2 + p_2 q_1 \in \mathbb{Z} \\ q' = q_1 \times q_2 \in \mathbb{Z}^+ \end{array} \right.$$

P₂: The product of two Rationals is Rational

Proof: Same previous method

$$x_1 \times x_2 = \frac{p'}{q'} \quad \left\{ \begin{array}{l} p' = p_1 \times p_2 \in \mathbb{Z} \\ q' = q_1 \times q_2 \in \mathbb{Z}^+ \end{array} \right. \\ \text{By assu. pthn}$$

P₃: The Inverse of non-zero Rational is Rational

Other Important proofs:

② I device p₂ → I device p

↳ This one means that If p² is an even number then p is even too.

→ proof by contradiction (p² is odd → p is odd)

② prove that $\sqrt{2}$ is irrational numbers.
prove by contradiction (assume that $\sqrt{2}$ is rational and so on...)

③ Interval of $\mathbb{R}$:

Definition:

A non empty subset of $\mathbb{R}$ is an Interval if verifies the following conditions

$$(\forall a, b \in I, \forall x \in \mathbb{R}, a \leq x \leq b) \Rightarrow (x \in I)$$

How to prove that I is an Interval?

$I = [0, 1]$ is Interval.

$$0 \in I \rightarrow I \neq \emptyset$$

$$\text{Let } a, b \in I, a < b$$

$$a \in I \Rightarrow 0 \leq a \leq 1$$

$$b \in I \Rightarrow 0 \leq b \leq 1$$

$$\text{Let } x \in \mathbb{R} \text{ s.t. } a \leq x \leq b$$

$$0 \leq a \leq x \leq b \leq 1 \Rightarrow 0 \leq x \leq 1 \Rightarrow x \in [0, 1]$$

I is Interval ✓

③ The density:

Theorem: $\mathbb{Q}$ is dense in $\mathbb{R}$.

$$\forall a, b \in \mathbb{R} \text{ s.t. } a < b, \exists r \in \mathbb{Q} \text{ s.t. } a < r < b$$

① The properties of $\mathbb{R}$:

① $(\mathbb{R}, +, \times)$ is commutative field

② $\mathbb{R}$ is an ordered set.

⑤ Upper and lower bounds:
 E is non empty subset of $\mathbb{R}$.

① E is bounded above:

$\exists \pi \in \mathbb{R}, \forall a \in E$ we have: $a \leq \pi$

↳ upper bound of E is π .

② E is bounded below:

$\exists m \in \mathbb{R}, \forall a \in E$ we have: $a \geq m$

↳ lower bound of E is m .

Quick Note:

① E is **bounded** if it is both $\rightarrow$ bounded below
 $\rightarrow$ bounded above

⑥ Supremum and Infimum:

E is non empty subset of $\mathbb{R}$.

① **Supremum** (E): is the least upper bound
 $m_0 = \sup(E) / \pi_0 \geq \pi$

② **Infimum** (E): is the greatest lower bound.

$m_0 = \inf(E) / m \leq m_0$

⑦ Maximum and Minimum:

① $\max E$: $\forall x \in E, x_0 \in E, x \leq x_0$ then $\max E = x_0$

② $\min E$: $\forall x \in E, x_0 \in E, x \geq x_0$ then $\min E = x_0$

Remark:

① If $\min E, \max E$ exists, they are unique

② If $\max$ exists, then $\sup E = \max E$

Proposition:

{ Let A and B are two non empty bounded subsets

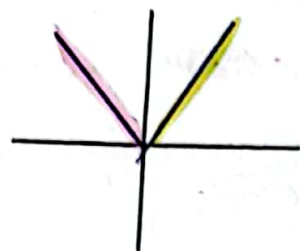
① $\sup(A \cup B) = \max(\sup A, \sup B)$

② $\inf(A \cup B) = \min(\inf A, \inf B)$

③ Absolute value:

Definition:

$|x| = \begin{cases} x & ; \text{if } x \geq 0 \\ -x & ; \text{if } x < 0 \end{cases}$



Proposition:

$\rightarrow$ ① $\sqrt{x^2} = |x| ; |x^n| = |x|^n$

$\rightarrow$ ② $|xy| = |x||y| ; |x-y| = |y-x|$

$\rightarrow$ ③ Triangle inequality: (check the proof)
 $i) |x+y| \leq |x| + |y|$
 $ii) |x-y| \geq |x| - |y|$

④ Integer part:

Definition: $\forall x \in \mathbb{R} \rightarrow [x] \leq x < [x] + 1$

$\rightarrow [x] - 1 < [x] \leq x$

Proposition:

$\rightarrow$ ① If $x \in \mathbb{Z}$ then: $[x] = x$.

exp: $[3] = 3$.

$\rightarrow$ ② $\forall x \in \mathbb{R}, n \in \mathbb{Z}$ then: $[x+n] = [x] + n$

exp:

$x = 2.3 \quad n = 3$

$[x+n] = [2.3+3] = 5$

$[x]+n = [2.3] + 3 = 5$